

Continual Learning with O-LoRA: Orthogonal Low-Rank Adapters for Sequential Tasks

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1. Introduction

Continual learning requires a model to sequentially acquire knowledge across multiple tasks without forgetting what it previously learned. A common challenge is **catastrophic forgetting**: adapting to a new task overwrites representations built for earlier ones. When parameter-efficient methods such as **LoRA** [1] are applied in this setting, the low-rank adapter matrices trained on successive tasks tend to occupy overlapping subspaces, causing destructive interference. **O-LoRA** [5] addresses this by imposing an orthogonality constraint on the A matrices of each new task adapter with respect to all previously learned ones, ensuring each task projects inputs into a distinct region of the low-rank latent space and limiting the interference that drives forgetting.

2. Dataset and Preprocessing

Task sequence (fixed order):

- **AG News** [6] – 4-class news topic classification
- **Amazon Polarity** [3] – binary sentiment classification
- **DBpedia 14** [6] – 14-class Wikipedia topic classification
- **Yahoo Answers Topics** [6] – 10-class question topics

Each task is trained on independently, with no replay of prior task data.

Instruction-tuning prompt format:

```
{task instruction}\n
Option: {comma-separated label names}\n
\n
Answer:
```

Only the input text is truncated when the sequence exceeds the maximum length. Cross-entropy loss is computed only on the label token; at evaluation the predicted label is the

6. References

[1] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[2] David Lopez-Paz and Marc’Aurelio Ranzato. Gradient episodic memory for continual learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. Defines BWT (Backward Transfer).

[3] Julian McAuley, Christopher Targett, Qinfeng Shi, and Anton van den Hengel. Image-based recommendations on styles and substitutes. In *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2015.

[4] Qwen Team. Qwen2.5 technical report. Technical report, Alibaba Group, 2024. URL <https://arxiv.org/abs/2412.15115>.

[5] Xiao Wang, Tianze Chen, Qiming Ge, Han Xia, Rong Bao, Rui Zheng, Qi Zhang, Tao Gui, and Xuanjing Huang. Orthogonal subspace learning for language model continual learning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

[6] Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

3. Models

Backbone: Qwen2.5-1.5B [4]. **LoRA adapters:** rank $r = 8$, scaling $\alpha = 32$, dropout 0.1, applied to the query and value projections of every attention layer. **Optimisation:** 1 epoch / task, learning rate 10^{-3} , batch size 8.

Compared methods:

- **IncLoRA** – interference-free upper bound: a separate adapter trained and frozen for each task (BWT = 0 by construction [2]).
- **O-LoRA** – a single cumulative adapter penalising overlap between the new task’s A matrix and all previously learned ones via an orthogonal regularisation term weighted by $\lambda_1 \in [0.2, 0.3]$.

Orthogonal loss. For task t with adapter $\Delta W_t = B_t A_t$:

$$\mathcal{L}_t = \mathcal{L}_{\text{CE}}(A_t, B_t) + \lambda_1 \sum_{i < t} \|A_t A_i^\top\|_F^2$$

The penalty drives the row-spaces of successive A matrices into mutually orthogonal directions. Crucially, **no constraint is applied to the B matrices**.

4. Results

IncLoRA baseline (no forgetting):		Impact of prompt structure:		
Task	Accuracy	Prompt format	Avg Acc	BWT
AG News	0.854	Old (no label options)	0.597	−0.344
Amazon Polarity	0.956	New (instruction + options)	0.799	−0.091
DBpedia 14	0.972			
Yahoo Answers Topics	0.640			

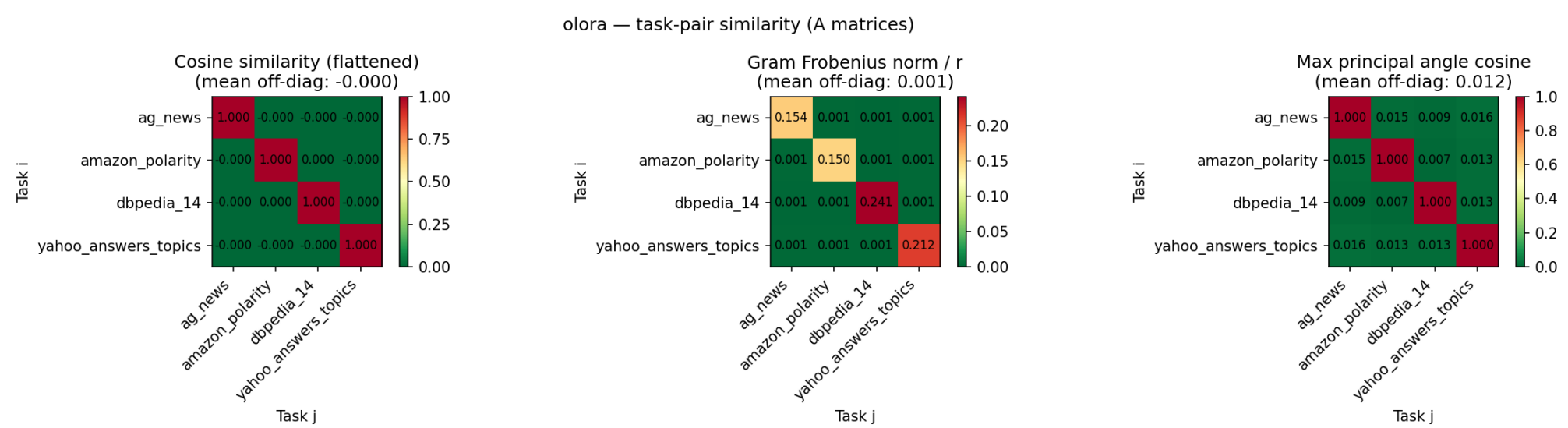
O-LoRA – summary metrics:

Run	Samples / task	λ_1	Avg Acc	BWT
Short	500/500/500/500	0.3	0.846	−0.032
2h	500/650/1850/1300	0.2	0.799	−0.091
Full (~50%)	2100/2700/7500/5300	0.3	0.654	−0.268

Per-task accuracy: initial → final

Task	Short	2h	Full (~50%)
AG News	0.878→0.825 (−6%)	0.842→0.747 (−11%)	0.887→0.494 (−44%)
Amazon Polarity	0.956→0.957 (~0%)	0.947→0.915 (−3%)	0.905→0.746 (−18%)
DBpedia 14	0.964→0.919 (−5%)	0.979→0.833 (−15%)	0.980→0.729 (−26%)
Yahoo Answers Topics	0.683→0.683 (—)	0.702→0.702 (—)	0.649→0.649 (—)

Avg accuracy and BWT both **degrade as the number of training samples grows**, even though more data improves *initial* per-task accuracy. The **instruction + options** prompt format yields +0.20 Avg Acc and +0.25 BWT – a larger effect than any hyperparameter choice.



Pairwise similarity of A matrices: cosine, Gram Frobenius norm $/r$, max principal angle cosine – all near-zero off-diagonal.

Orthogonality analysis. All three metrics show near-zero similarity across A even with $\lambda_1 \in [0.2, 0.3]$ – the constraint is easy to satisfy. However B matrices, unconstrained by the loss, share ~ 30% of their column-space directions (principal-angle cosine ~ 0.29) and grow in magnitude with training – plausibly the primary driver of forgetting at higher scales.

5. Conclusion

- O-LoRA’s regulariser keeps A matrices **geometrically separated** with mild penalties; translating this into reduced forgetting is **highly sensitive to data scale** – near-zero BWT at 500 samples/task, sharp collapse (−0.268) at ~50% of the paper’s scale.
- The unregularised B matrices grow in magnitude and share ~ 30% of their column-space directions across tasks – a likely primary driver of forgetting at scale.
- **Prompt structure matters more than hyperparameters:** explicitly listing label names yields +0.20 accuracy and +0.25 BWT.